

WEEK 7 ASSIGNMENT
H524 Introduction to Biostatistics - Fall 2025

Total Points: 10

Due: Sunday 11:59 PM

Part A: Multiple Choice Questions (5 points)

Answer the following 5 questions on nonparametric methods. Each question is worth 1 point.

Question 1: When to Use Nonparametric Tests (1 point)

Which of the following is the BEST reason to use a nonparametric test instead of a parametric test?

- A. Nonparametric tests are always more powerful than parametric tests
- B. The data do not meet the assumptions required for parametric tests (e.g., normality)
- C. Nonparametric tests require larger sample sizes
- D. Parametric tests cannot be used with continuous data

Question 2: Sign Test vs. t-test (1 point)

A researcher wants to test whether a new medication reduces pain scores in patients. Pre- and post-treatment pain scores are measured for 15 patients. The pain score differences are highly skewed. Which test is most appropriate?

- A. Paired t-test
- B. Sign test
- C. Independent samples t-test
- D. ANOVA

Question 3: Mann-Whitney U Test (1 point)

The Mann-Whitney U test is the nonparametric alternative to which parametric test?

- A. Paired t-test
- B. One-way ANOVA
- C. Independent samples t-test (two-sample t-test)
- D. Chi-square test

Question 4: Kruskal-Wallis Test (1 point)

A study compares median recovery times across four different treatment groups. The recovery time data are highly right-skewed. Which test is most appropriate?

- A. One-way ANOVA
- B. Kruskal-Wallis test
- C. Chi-square test
- D. Two-sample t-test

Question 5: Ranking Data (1 point)

In the Wilcoxon rank-sum test (Mann-Whitney U), how are tied values handled?

- A. Ties are broken randomly
- B. Tied values receive the average of the ranks they would have occupied
- C. Tied values are excluded from the analysis
- D. Ties are not allowed in nonparametric tests

Part B: Group Project Check-In - Team Plan (5 points)

GROUP PROJECT BEGINS!

Read First: See `Group_Project_Overview.pdf` posted on Canvas for complete project details.

Your Assignment

Group Leader submits a team plan including:

1. Team Roster

- Names of all group members
- Contact information (email, phone)

2. Communication Plan

- How will you communicate? (email, text, Slack, Discord, etc.)
- Preferred meeting format (in-person, Zoom, hybrid)

3. Meeting Schedule

- At least 2-3 scheduled meeting times for Weeks 8-10
- Include dates and times

4. Division of Responsibilities

- Who will handle initial tasks:
 - Read data dictionary
 - Set up R and load dataset
 - Contact AI tools (Claude, ChatGPT)
 - Use Copilot in RStudio
 - Draft presentation outline
 - Write report sections

5. Preliminary Timeline

- When will you complete data analysis?
- When will you finish AI comparison?
- When will you draft presentation?
- When will you finalize report?

Submission Format

Submit as: Text entry OR file upload (PDF/Word)

Length: 1-2 pages

Grading: Completion-based (5 points for submitting reasonable plan)

Notes

- **Only the group leader submits** - all team members get the same grade
- Check Canvas to see your group number and dataset assignment
- Download your dataset from: Canvas → Files → Group Project Data → **group_X** folder
- See data dictionary in Canvas Files for variable descriptions
- This is just a PLAN - you can adjust as you go!